

Northeastern University

Tourism and Jobs in Maine: Evaluating Acadia National Park's Impact on Local Employment

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MISM 6200 — Introduction to Business Analytics

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Executive Summary

This project analyzes how seasonal tourism at Acadia National Park from January 2015 through December 2024 drives predictable patterns in local employment. Starting from the intuitive observation that national parks typically attract more visitors during warmer months, I used a structured analytical process to quantify this relationship and evaluate its economic implications. Ten years of National Oceanic and Atmospheric Administration (NOAA) climate data show a highly regular seasonal temperature cycle, with average daily temperatures rising from ~25°F in winter to ~70°F in summer. The environmental pattern is mirrored exceptionally well in monthly visitation trends taken directly from the National Park Service (NPS), as Acadia averages fewer than 20,000 visitors in winter months but exceeds 700,000 in July and August. Linear and logistic regression models confirm that temperature alone explains a very large share of visitation variation (R^2 ranging from 0.84 to 0.94), demonstrating that seasonal climate is the dominant driver of park traffic.

I then linked the annual visitation cycle to detailed employment data from the Bureau of Labor Statistics (BLS) across four Maine counties, revealing a striking contrast. Hancock County, located directly around Acadia, shows dramatic annual swings in employment across multiple industries, while inland counties remain comparatively stable. Aggregated employment in Hancock rises by an average of 6,660 jobs from winter to peak season, a 30% increase that is clearly tied to the summer surge in visitors. Disaggregated industry analysis reveals that Leisure & Hospitality absorbs 4,293 of these positions—64.5% of all seasonal job growth—and its employment trend curve closely matches the monthly tourist inflow. Additional sectors such as Trade/Transportation/Utilities and Professional Services show noticeable but smaller seasonal increases, while most non-tourism industries remain stable. Regression slopes quantify this relationship; for Leisure & Hospitality, approximately 178 additional visitors correspond to one additional job, highlighting both sensitivity and economic reliance on tourism. Even more striking, the model's R^2 of 98% indicates that monthly visitor volume almost perfectly explains employment fluctuations in this sector, a level of fit that is rare but economically justified given the direct, compounding link between tourism demand and staffing needs.

Overall, the evidence demonstrates that Acadia functions as a powerful and predictable economic engine whose influence is localized, quantifiable, and strongly seasonal. For local businesses, these insights enable proactive planning around staffing, resource allocation, inventory, service capacity, and seasonal investment strategies. By recognizing how tightly temperature, visitation, and employment are linked, organizations can forecast labor needs months in advance, reduce operational uncertainty, and more effectively capitalize on Acadia's reliable summer-driven economic cycle.

Appendix A — Data Sources & Core Variables

This project integrates three primary datasets covering January 2015 through December 2024: Acadia National Park visitation, NOAA climate observations, and Bureau of Labor Statistics employment data. These sources were cleaned, merged, and transformed using Excel and R to create a unified analytical framework linking weather patterns, seasonal visitation, and county-level employment outcomes across Maine.

1. Weather Data (weather-data.csv)

Source: National Oceanic and Atmospheric Administration (NOAA)

Description: Daily temperature observations for the Acadia region, aggregated to monthly values.

Core Variables Used:

- Date — Daily date stamp (converted to monthly via year(Date) and month(Date)).
- TMAX, TMIN, TAVG — Daily high, low, and average temperatures (°F).
- AvgHigh / AvgTemp / AvgLow — Monthly average high, average, and low temperatures, used to model and visualize Acadia’s climate cycle.
- Month, Year — Extracted temporal identifiers for joining and grouping.

2. Visitation Data (acadia-visitors.csv)

Source: National Park Service (NPS Visitor Use Statistics)

Description: Annual dataset containing monthly recreation visitor counts to Acadia National Park.

Core Variables Used:

- Year — Calendar year of the record.
- JAN–DEC — Monthly visitor counts reshaped into long format.
- Visitors — Monthly visitation counts used throughout modeling.
- AvgVisitors — Monthly averages across all years (2015–2024) computed to smooth noise and highlight typical seasonal patterns.

3. Employment Data (employment_maine_4counties_allindustries.csv)

Source: U.S. Bureau of Labor Statistics (BLS)

Description: Monthly employment totals for all NAICS industries across four Maine counties: Hancock, Penobscot, Kennebec, and Cumberland.

Core Variables Used:

- Area — County name (e.g., “Hancock County, Maine”).
- Industry — Industry classification (cleaned in the code as IndustryClean).
- Employment — Monthly employment count per industry.
- Date — Constructed from Year and Month for time-series alignment.
- Tier / Approx_Distance_miles — Indicators used to categorize control counties by distance from Acadia.

Appendix B — Visualizations

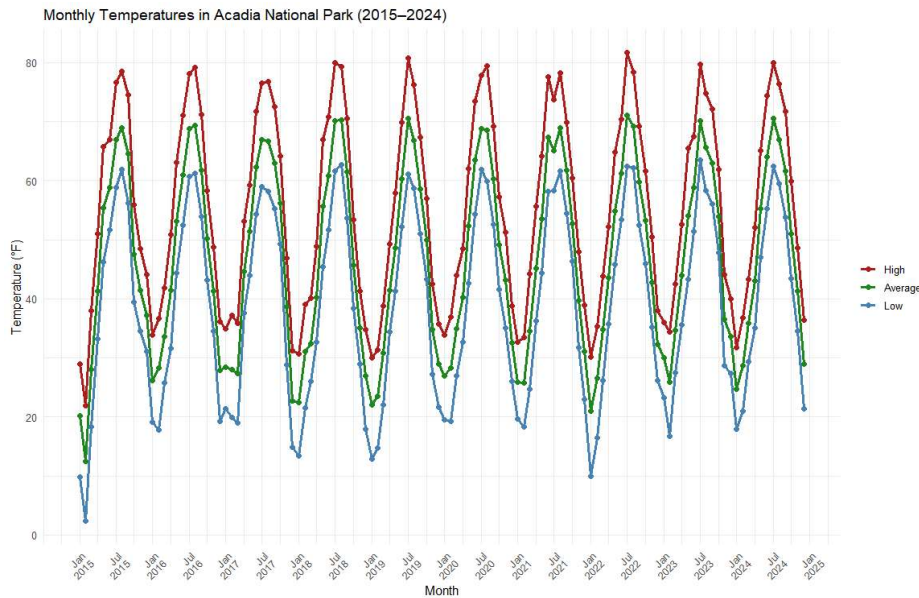


Figure 1: Monthly Temperatures in Acadia National Park

This line graph shows the monthly high, average, and low temperatures in Acadia from 2015–2024, highlighting a highly consistent seasonal cycle where temperatures peak in July–August and reach their lowest in January–February.

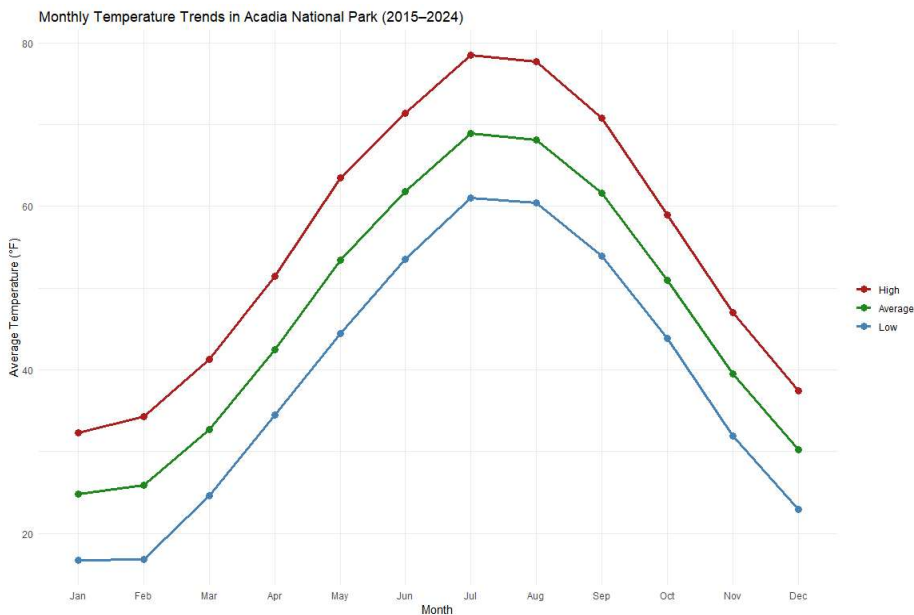


Figure 2: Monthly Temperature Trends in Acadia National Park

This line graph averages the monthly high, average, and low temperatures across all years from 2015–2024, smoothing out year-to-year noise to clearly show the typical seasonal temperature pattern in Acadia, providing a cleaner view of the climate cycle that drives seasonal visitation.

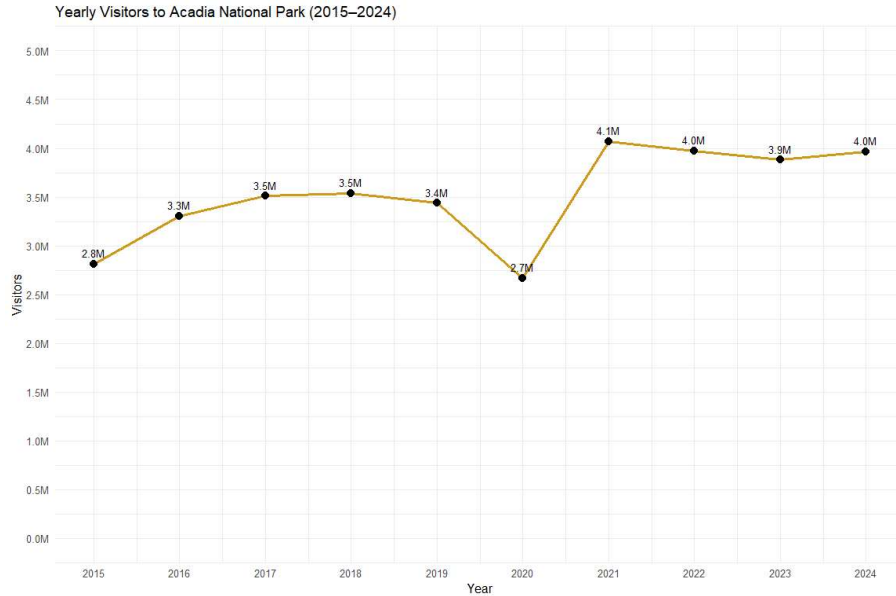


Figure 3: Yearly Visitors to Acadia National Park

This line graph displays annual visitor totals to Acadia from 2015–2024, showing gradual growth leading up to 2019, a notable dip in 2020, and a strong recovery in the years that follow. It provides a big-picture view of how overall demand for the park has changed over the decade.

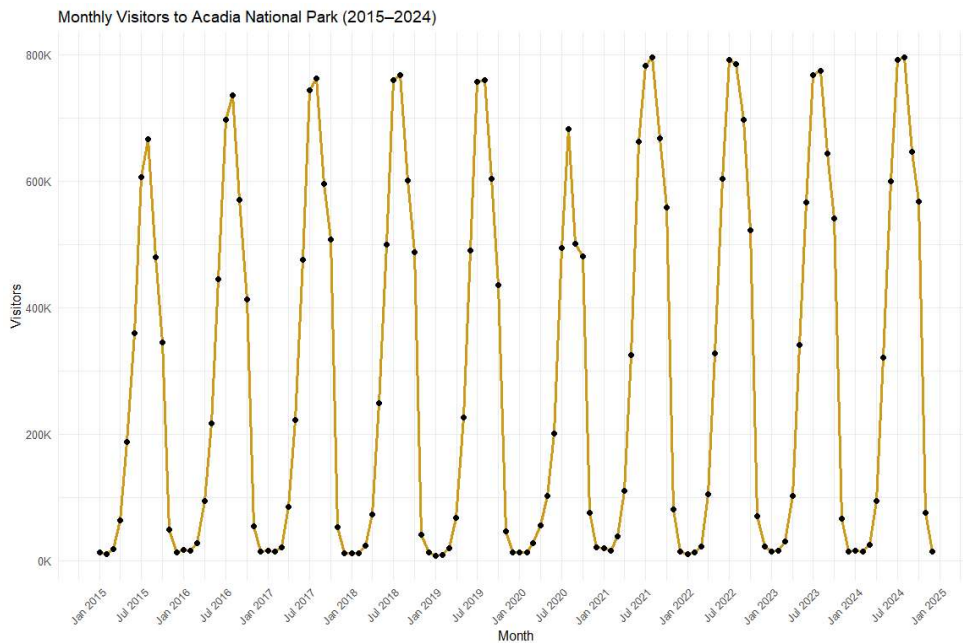


Figure 4: Monthly Visitors to Acadia National Park

This line graph shows monthly visitor counts to Acadia from 2015–2024, with steep summer peaks and quiet winter months that closely mirror the seasonal temperature pattern. It highlights how visitor activity rises and falls almost in sync with the park’s climate cycle.

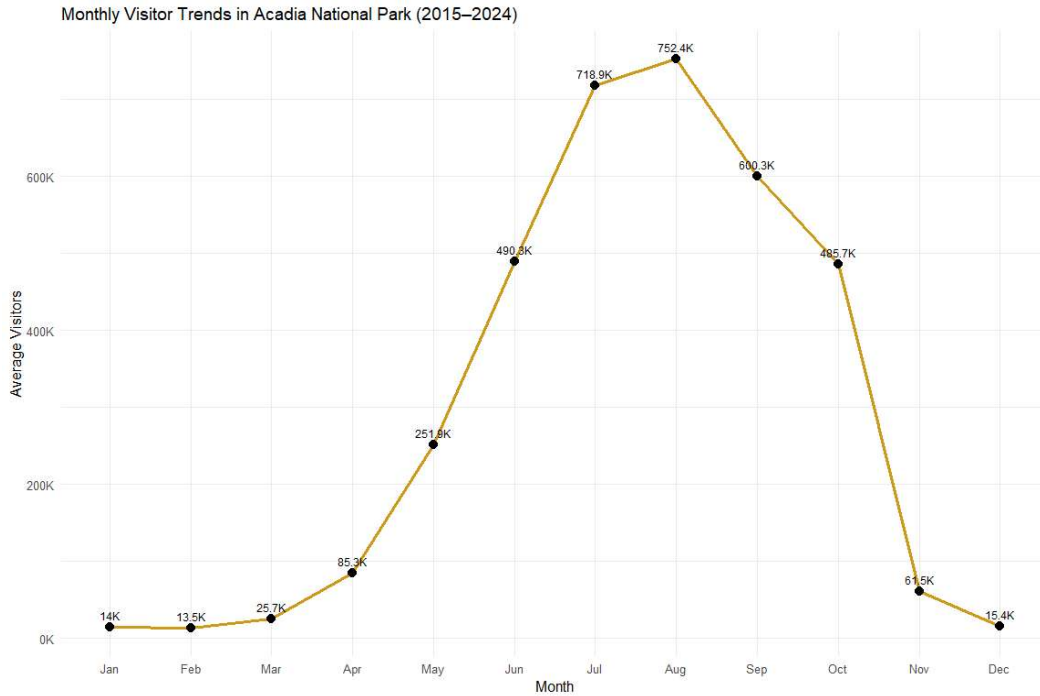


Figure 5: Monthly Visitor Trends in Acadia National Park

As with the temperature data, this line graph averages visitor counts by month across the ten-year period to clearly show the typical seasonal pattern.

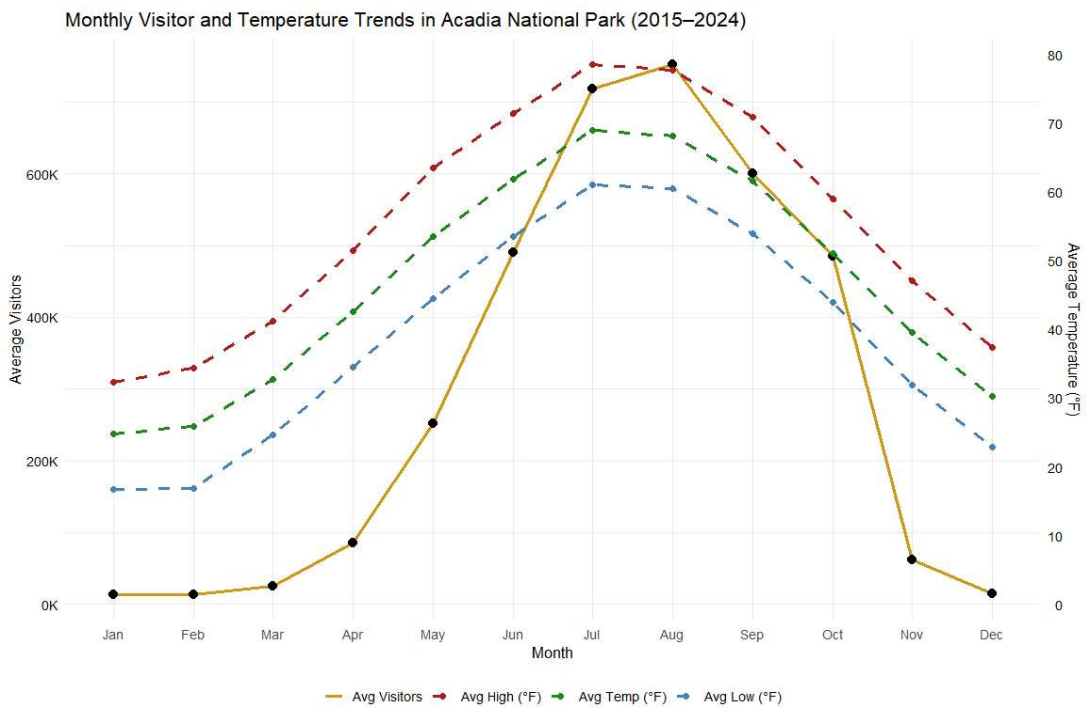


Figure 6: Monthly Visitor and Temperature Trends in Acadia National Park

This line graph combines the previous temperature and visitation trend graphs into a single view, making it easy to see how closely the two seasonal patterns align throughout the year.

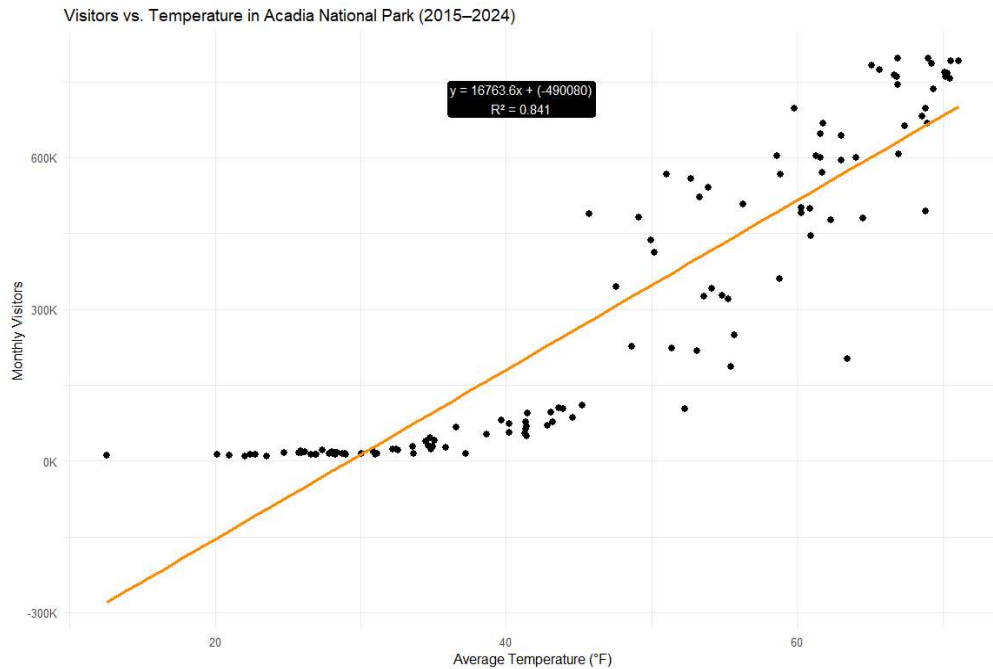


Figure 7: Visitors vs. Temperature in Acadia National Park

This scatter plot shows the relationship between average monthly temperature and monthly visitors, with a linear regression indicating a fairly strong positive correlation ($R^2 = 0.84$). For every additional degree Fahrenheit, Acadia receives roughly 16,764 additional visitors.

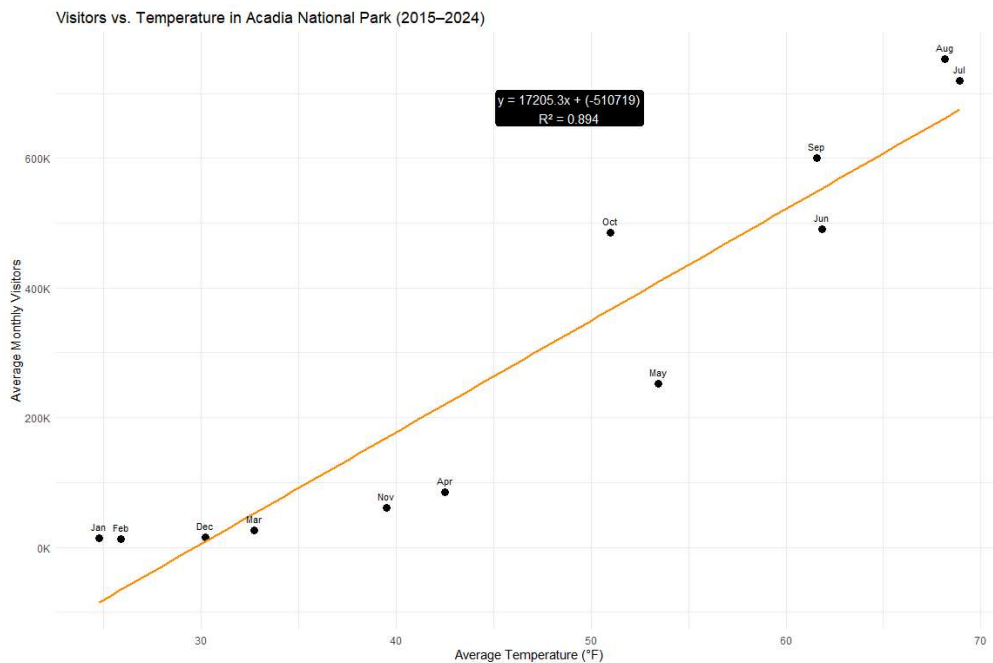


Figure 8: Visitors vs. Temperature in Acadia National Park — Monthly Averages

Building on the previous scatter plot, this scatter plot averages each month across all ten years to reduce noise and produces a stronger linear model ($R^2 = 0.89$). For every additional degree Fahrenheit, Acadia receives roughly 17,205 additional visitors.

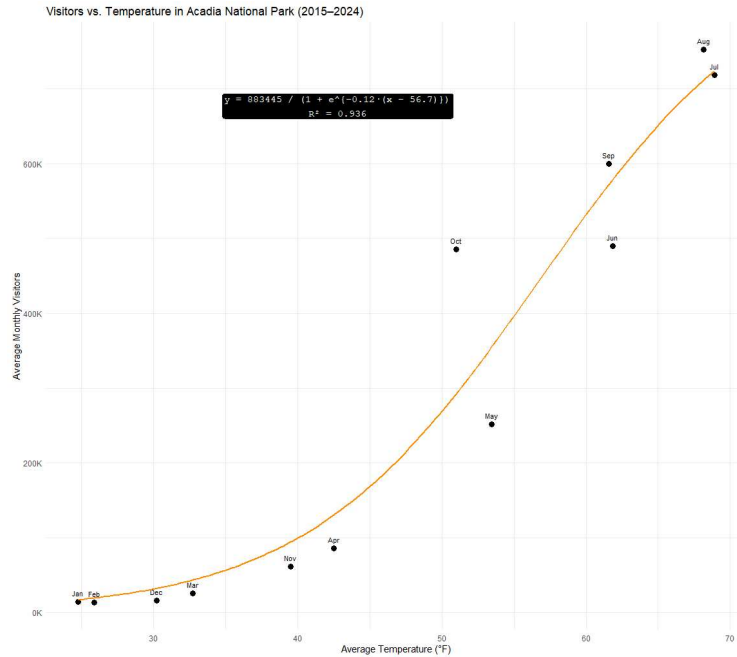


Figure 9: Visitors vs. Temperature in Acadia National Park — Monthly Averages, Logistic Curve

This scatter plot extends the previous analysis by using a logistic regression instead of a linear model, which better captures the slowing growth in visitation at very low and very high temperatures. The model achieves a strong fit ($R^2 = 0.94$), and its curve shows that visitor numbers accelerate rapidly as temperatures rise toward the mid-50s, then level off as the park approaches its natural capacity in peak summer months.

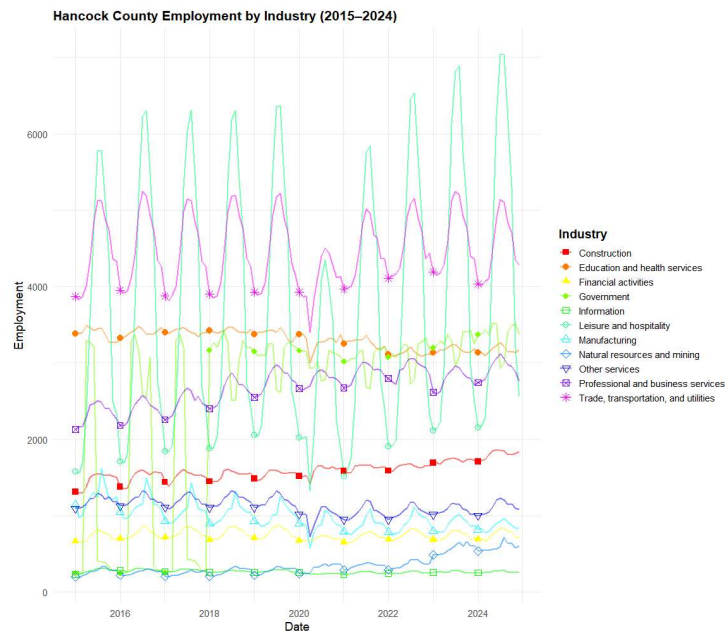


Figure 10: Hancock County Employment by Industry

This line graph shows monthly employment levels across major industries in Hancock County from 2015–2024, revealing clear seasonal patterns in several different sectors.

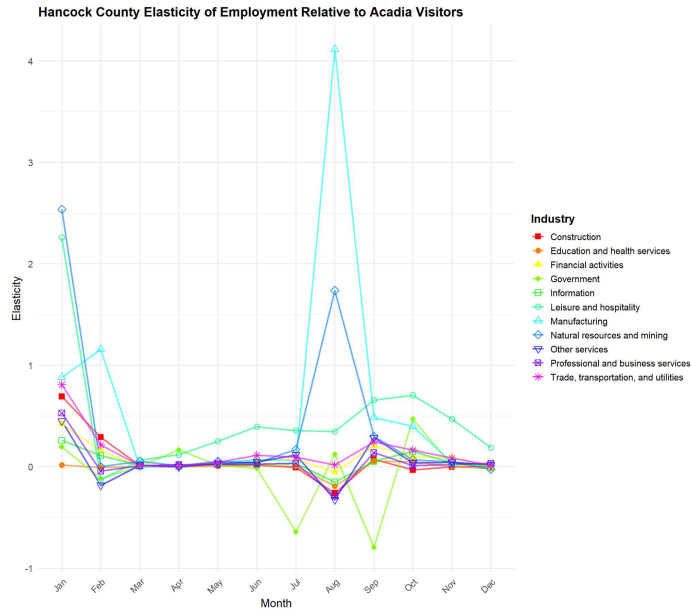


Figure 11: Hancock County Elasticity of Employment Relative to Acadia Visitors

This line graph shows the monthly average elasticity of employment relative to Acadia visitors in Hancock County, where elasticity measures the percent change in employment for a percent change in visitation. However, because elasticity becomes unstable when monthly employment levels or visitor counts are very small, especially in winter, the trends appear noisy and are difficult to interpret cleanly from this plot alone.

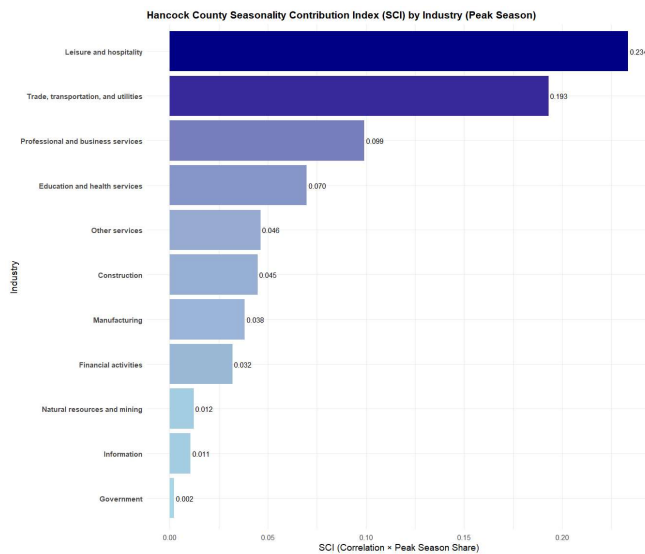


Figure 12: Hancock County Seasonality Contribution Index (SCI) by Industry (Peak Season)

This bar graph shows the Seasonality Contribution Index for each industry in Hancock County, ranking how much each sector contributes to peak-season employment fluctuation based on its correlation with park visitation trends multiplied by its relative share of employment during July–September. Based on the results, Leisure & Hospitality, Trade/Transportation/Utilities, and Professional Services are defined as our three most visitor-sensitive industries.

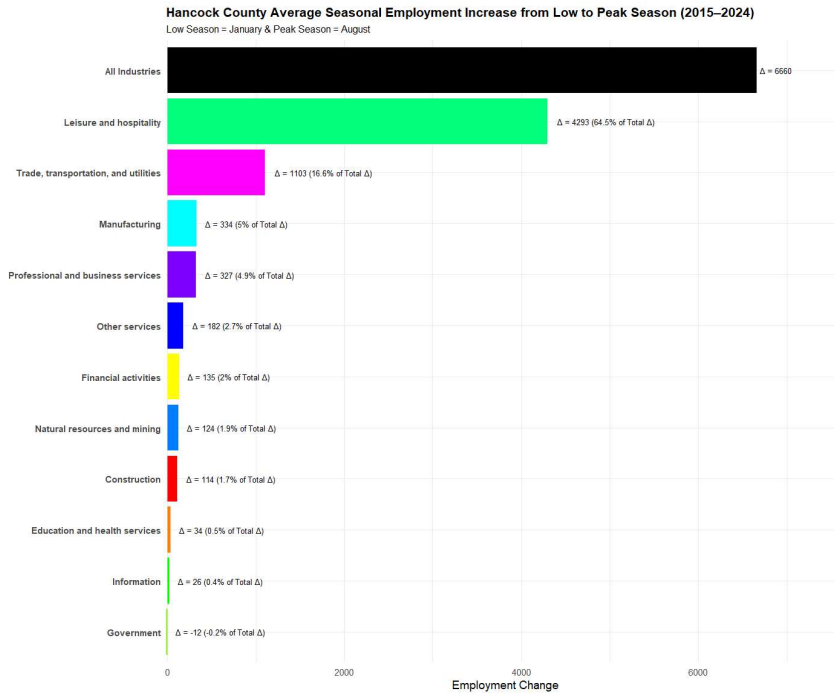


Figure 13: Hancock County Average Seasonal Employment Increase from Low to Peak Season

This bar graph shows how much each industry in Hancock County increases its employment from the winter low (January) to the summer peak (August), highlighting both the size of the seasonal surge and the share contributed by each sector.

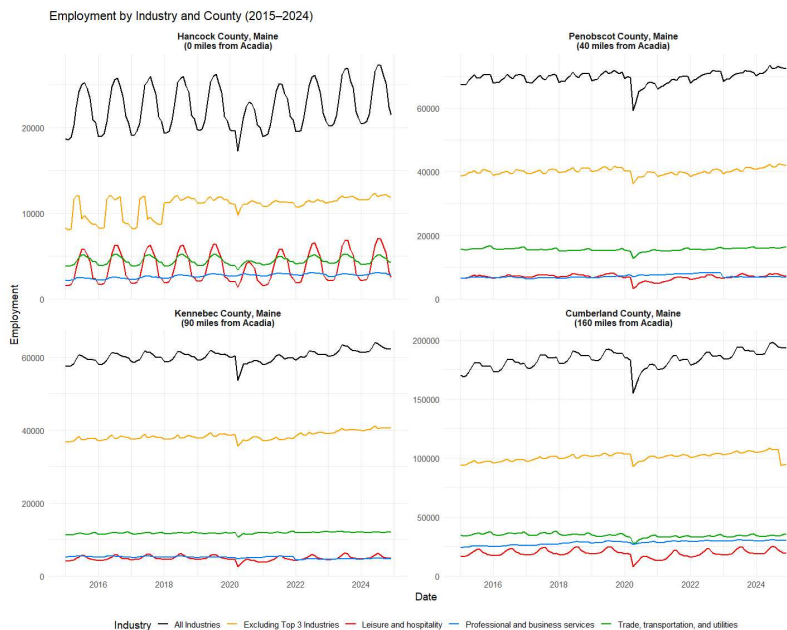


Figure 14: Employment by Industry and County

This faceted line graph plots monthly employment from January 2015 to December 2024 for the top three visitor-sensitive industries, showing Hancock County in the top-left panel alongside three nearby counties used as controls.

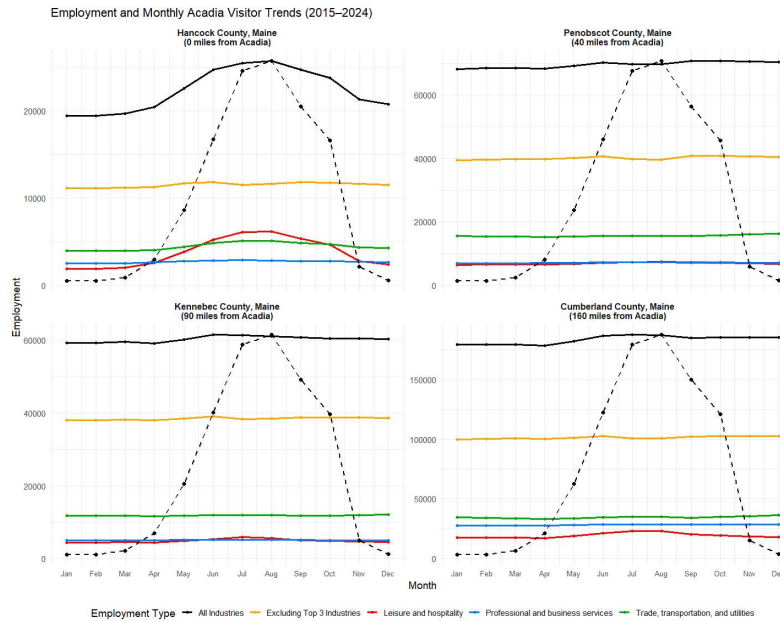


Figure 15: Employment and Monthly Acadia Visitor Trends

This faceted line graph takes monthly averages of the previous employment plot and overlays Acadia’s visitor curve to allow direct comparison of their seasonal trends.

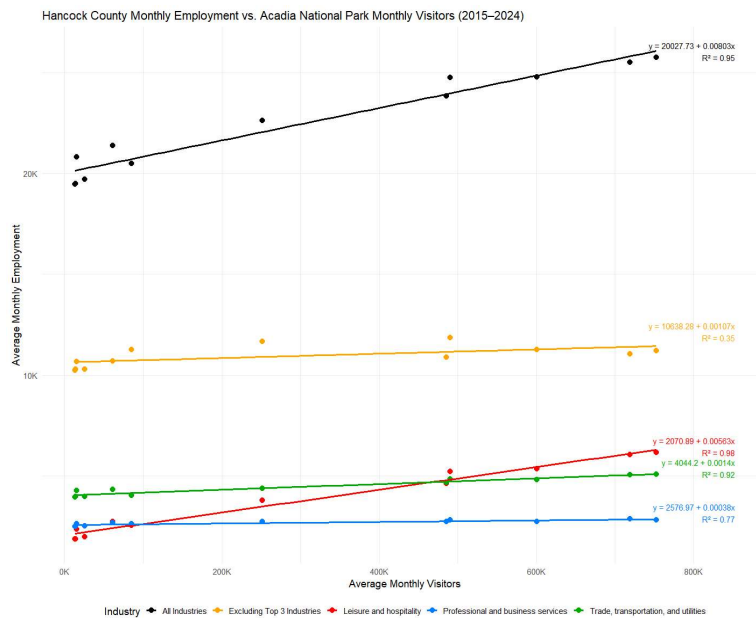


Figure 16: Hancock County Monthly Employment vs. Acadia National Park Monthly Visitors

This scatter plot displays five linear regression lines—one for all industries, one excluding the top three tourism-sensitive sectors, and one for each of the top three industries—showing how average monthly employment relates to average monthly visitors in Hancock County. The coefficients and R^2 values are shown for each line, with Leisure & Hospitality standing out for its exceptionally strong fit ($R^2 = 0.98$).

Employment Sensitivity to Park Visitation		
Slope and Inverse Slope of Employment ~ Visitors Regression		
Industry	Slope	Inverse Slope
All Industries	0.00803	125
Excluding Top 3 Industries	0.00107	936
Leisure and hospitality	0.00563	178
Professional and business services	0.00038	2,638
Trade, transportation, and utilities	0.00140	712

Figure 17: Employment Sensitivity to Park Visitation

This table better interprets the regression coefficients from the previous plot (initially expressed as jobs per visitor) by taking their inverses, converting each slope into the number of visitors associated with one additional job in each industry; it directly quantifies how increases in Acadia visitation translate into employment growth in Hancock County.

Appendix C — Full Reproducible Code

```
# MISM6200 Final Project
# Tourism and Jobs in Maine: Evaluating Acadia National Park's Impact on Local Employment
# Patrick Medek

library(tidyverse)
library(lubridate)
library(scales)
library(glue)
library(ggpmisc)
library(gt)

# === 1. LOAD DATA ===

weather <- read_csv("C:/Users/patri/Documents/Northeastern/Intro to Business Analytics/Final Project/Data/weather-data.csv")
visitors <- read_csv("C:/Users/patri/Documents/Northeastern/Intro to Business Analytics/Final Project/Data/acadia-visitors.csv")
employment <- read_csv("C:/Users/patri/Documents/Northeastern/Intro to Business Analytics/Final Project/Data/Employment/employment_maine_4counties_allindustries.csv")

# === 2. CLEAN DATA ===

# --- WEATHER ---
if ("DATE" %in% names(weather)) names(weather)[names(weather) == "DATE"] <- "Date"

weather <- weather %>%
  mutate(
    Date = as.Date(Date, format = "%m/%d/%Y"),
    Year = year(Date),
    Month = month(Date, label = TRUE, abbr = TRUE)
  ) %>%
  select(Year, Month, Date, TMAX, TMIN, TAVG) %>%
  mutate(across(c(TMAX, TMIN, TAVG), as.numeric))

weather_monthly <- weather %>%
  mutate(
    Year = year(Date),
    MonthNum = month(Date),
    MonthLabel = month(Date, label = TRUE, abbr = TRUE)
  ) %>%
  group_by(Year, MonthNum, MonthLabel) %>%
  summarise(
    AvgTemp = mean(TAVG, na.rm = TRUE),
    AvgTMax = mean(TMAX, na.rm = TRUE),
    AvgTMin = mean(TMIN, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(Year, MonthNum) %>%
  mutate(
    MonthLabelFull = paste0(MonthLabel, " ", substr(Year, 3, 4))
  )

weather_monthly_avg <- weather %>%
  group_by(Month) %>%
  summarise(
    AvgHigh = mean(TMAX, na.rm = TRUE),
    AvgLow = mean(TMIN, na.rm = TRUE),
    AvgTemp = mean(TAVG, na.rm = TRUE)
  )

# --- VISITORS ---
visitors_clean <- visitors %>%
  rename(Year = YEAR) %>%
  pivot_longer(
    cols = c("JAN", "FEB", "MAR", "APR", "MAY", "JUN", "JUL", "AUG", "SEP", "OCT", "NOV", "DEC"),
    names_to = "Month",
    values_to = "Visitors"
  ) %>%
  mutate(
    Month = factor(
      Month,
      levels = c("JAN", "FEB", "MAR", "APR", "MAY", "JUN",
                  "JUL", "AUG", "SEP", "OCT", "NOV", "DEC"),
      labels = month.abb,
      ordered = TRUE
    )
  )
```

```

)

visitors_monthly_avg <- visitors_clean %>%
  group_by(Month) %>%
  summarise(AvgVisitors = mean(Visitors, na.rm = TRUE))

# --- EMPLOYMENT ---
employment <- employment %>%
  mutate(
    MonthNum = match(Month, month.abb),
    Date = ymd(paste(Year, MonthNum, "01", sep = "-")),
    IndustryClean = str_replace(Industry, "^([0-9]+\\s+)", "")
  )

all_jobs <- employment %>%
  filter(
    Ownership == "Total Covered",
    IndustryClean == "Total, all industries"
  ) %>%
  transmute(
    Area, Tier, Approx_Distance_miles, Date,
    Industry = "All Jobs",
    Employment
  )

government <- employment %>%
  filter(
    Ownership %in% c("Federal Government", "State Government", "Local Government")
  ) %>%
  group_by(Area, Tier, Approx_Distance_miles, Date) %>%
  summarise(
    Industry = "Government",
    Employment = sum(Employment),
    .groups = "drop"
  )

subtotal_industries <- c(
  "Total, all industries",
  "Goods-producing",
  "Service-providing"
)

private_subtotals <- c(
  "Private",
  "Private, goods-producing",
  "Private, service-providing"
)

private_leaf <- employment %>%
  filter(
    Ownership == "Private",
    !IndustryClean %in% subtotal_industries,
    !IndustryClean %in% private_subtotals,
    !IndustryClean %in% "Unclassified"
  ) %>%
  transmute(
    Area, Tier, Approx_Distance_miles, Date,
    Industry = IndustryClean,
    Employment
  )

employment <- bind_rows(
  all_jobs,
  government,
  private_leaf
)

employment_monthly <- employment %>%
  mutate(
    MonthNum = month(Date),
    MonthLabel = month.abb[MonthNum]
  )

emp_monthly_avg <- employment_monthly %>%
  group_by(Area, Tier, Approx_Distance_miles, Industry, MonthNum, MonthLabel) %>%
  summarize(
    AvgEmployment = round(mean(Employment, na.rm = TRUE)),
    .groups = "drop"
  )

```

```

)

# === 3. WEATHER GRAPHS ===

# Monthly Temperatures in Acadia National Park (2015-2024)
weather_monthly_actual <- weather %>%
  group_by(Year, Month) %>%
  summarise(
    AvgHigh = mean(TMAX, na.rm = TRUE),
    AvgLow = mean(TMIN, na.rm = TRUE),
    AvgTemp = mean(TAVG, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    MonthNum = match(Month, month.abb),
    Date = as.Date(paste(Year, MonthNum, "01", sep = "-"))
  )

weather_long_actual <- weather_monthly_actual %>%
  pivot_longer(
    cols = c(AvgHigh, AvgTemp, AvgLow),
    names_to = "Type",
    values_to = "Temp"
  ) %>%
  mutate(Type = factor(Type, levels = c("AvgHigh", "AvgTemp", "AvgLow")))

ggplot(weather_long_actual, aes(x = Date, y = Temp, color = Type, group = Type)) +
  geom_line(size = 1.1) +
  geom_point(size = 2) +
  scale_color_manual(
    values = c("AvgHigh" = "firebrick", "AvgTemp" = "forestgreen", "AvgLow" = "steelblue"),
    labels = c("High", "Average", "Low")
  ) +
  scale_x_date(
    date_breaks = "6 months",
    date_labels = "%b\n%Y"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.title = element_blank()
  ) +
  labs(
    title = "Monthly Temperatures in Acadia National Park (2015-2024)",
    x = "Month",
    y = "Temperature (°F)",
    color = "Temperature Type"
  )

# Monthly Temperature Trends in Acadia National Park (2015-2024)
ggplot(weather_monthly_avg, aes(x = Month, group = 1)) +
  geom_line(aes(y = AvgHigh, color = factor("High", levels = c("High", "Average", "Low"))), size = 1.2) +
  geom_line(aes(y = AvgTemp, color = factor("Average", levels = c("High", "Average", "Low"))), size = 1.2) +
  geom_line(aes(y = AvgLow, color = factor("Low", levels = c("High", "Average", "Low"))), size = 1.2) +
  geom_point(aes(y = AvgHigh, color = factor("High", levels = c("High", "Average", "Low"))), size = 3) +
  geom_point(aes(y = AvgTemp, color = factor("Average", levels = c("High", "Average", "Low"))), size = 3) +
  geom_point(aes(y = AvgLow, color = factor("Low", levels = c("High", "Average", "Low"))), size = 3) +
  scale_color_manual(
    values = c("High" = "firebrick", "Average" = "forestgreen", "Low" = "steelblue"),
    breaks = c("High", "Average", "Low"),
    name = NULL
  ) +
  theme_minimal() +
  labs(
    title = "Monthly Temperature Trends in Acadia National Park (2015-2024)",
    y = "Average Temperature (°F)",
    x = "Month"
  )

# === 4. VISITOR GRAPHS ===

# Yearly Visitors to Acadia National Park (2015-2024)
visitors_clean %>%
  distinct(Year, TOTAL, .keep_all = FALSE) %>%
  ggplot(aes(x = Year, y = TOTAL)) +
  geom_line(color = "goldenrod3", size = 1.2) +

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geom_point(color = "black", size = 3) +
geom_text(
  aes(label = label_number(scale = 1e-6, suffix = "M", accuracy = 0.1)(TOTAL)),
  vjust = -0.8,
  color = "black",
  size = 3.5
) +
scale_x_continuous(breaks = 2015:2024) +
scale_y_continuous(
  limits = c(0, 5000000),
  breaks = seq(0, 5000000, 500000),
  labels = label_number(scale = 1e-6, suffix = "M", accuracy = 0.1)
) +
theme_minimal(base_size = 12) +
labs(
  title = "Yearly Visitors to Acadia National Park (2015-2024)",
  x = "Year",
  y = "Visitors"
)

# Monthly Visitors to Acadia National Park (2015-2024)
visitors_clean %>%
mutate(
  Date = as.Date(paste(Year, match(Month, month.abb), "01", sep = "-"))
) %>%
ggplot(aes(x = Date, y = Visitors)) +
geom_line(color = "goldenrod3", size = 1.2) +
geom_point(color = "black", size = 2) +
scale_y_continuous(
  labels = label_number(scale = 1e-3, suffix = "K")
) +
scale_x_date(
  date_breaks = "6 months",
  date_labels = "%b %Y",
) +
theme_minimal(base_size = 12) +
theme(
  axis.text.x = element_text(angle = 45, hjust = 1)
) +
labs(
  title = "Monthly Visitors to Acadia National Park (2015-2024)",
  x = "Month",
  y = "Visitors"
)

# Monthly Visitor Trends in Acadia National Park (2015-2024)
ggplot(visitors_monthly_avg, aes(x = Month, y = AvgVisitors, group = 1)) +
geom_line(color = "goldenrod3", size = 1.2) +
geom_point(color = "black", size = 3) +
geom_text(
  aes(label = paste0(round(AvgVisitors / 1000, 1), "K")),
  vjust = -0.8, size = 3.2, color = "black"
) +
scale_y_continuous(
  labels = label_number(scale = 1/1000, suffix = "K")
) +
theme_minimal(base_size = 12) +
labs(
  title = "Monthly Visitor Trends in Acadia National Park (2015-2024)",
  y = "Average Visitors",
  x = "Month"
)

# === 5. COMBINED WEATHER + VISITORS ===

weather_monthly_avg <- weather_monthly_avg %>%
mutate(Month = factor(Month, levels = month.abb, ordered = FALSE))
visitors_monthly_avg <- visitors_monthly_avg %>%
mutate(Month = factor(Month, levels = month.abb, ordered = FALSE))

combined_monthly <- left_join(visitors_monthly_avg, weather_monthly_avg, by = "Month")

# Monthly Visitor and Temperature Trends in Acadia National Park (2015-2024)
scale_factor <- max(combined_monthly$AvgVisitors, na.rm = TRUE) / max(combined_monthly$AvgHigh, na.rm = TRUE)

ggplot(combined_monthly, aes(x = Month, group = 1)) +
geom_line(aes(y = AvgVisitors, color = "Avg Visitors"), size = 1.3) +

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geom_point(aes(y = AvgVisitors), color = "black", size = 3) +
geom_line(aes(y = AvgHigh * scale_factor, color = "Avg High (°F)"), size = 1.2, linetype = "dashed") +
geom_line(aes(y = AvgTemp * scale_factor, color = "Avg Temp (°F)"), size = 1.2, linetype = "dashed") +
geom_line(aes(y = AvgLow * scale_factor, color = "Avg Low (°F)"), size = 1.2, linetype = "dashed") +
geom_point(aes(y = AvgHigh * scale_factor, color = "Avg High (°F)"), size = 2) +
geom_point(aes(y = AvgTemp * scale_factor, color = "Avg Temp (°F)"), size = 2) +
geom_point(aes(y = AvgLow * scale_factor, color = "Avg Low (°F)"), size = 2) +

scale_y_continuous(
  name = "Average Visitors",
  labels = label_number(scale = 1e-3, suffix = "K"),
  sec.axis = sec_axis(~ . / scale_factor,
    name = "Average Temperature (°F)",
    breaks = seq(0, 90, 10))
) +

scale_color_manual(
  values = c(
    "Avg Visitors" = "goldenrod3",
    "Avg High (°F)" = "firebrick",
    "Avg Temp (°F)" = "forestgreen",
    "Avg Low (°F)" = "steelblue"
  ),
  breaks = c("Avg Visitors", "Avg High (°F)", "Avg Temp (°F)", "Avg Low (°F)"),
  name = NULL
) +
theme_minimal(base_size = 13) +
theme(
  axis.title.y.left = element_text(color = "black", size = 12),
  axis.title.y.right = element_text(color = "black", size = 12),
  axis.text.y.left = element_text(color = "black"),
  axis.text.y.right = element_text(color = "black"),
  legend.position = "bottom"
) +
labs(
  title = "Monthly Visitor and Temperature Trends in Acadia National Park (2015-2024)",
  x = "Month"
)

# Visitors vs. Temperature in Acadia National Park (2015-2024)
combined_monthly_all <- weather_monthly %>%
  left_join(
    visitors_clean,
    by = c("Year", "MonthLabel" = "Month")
  ) %>%
  select(Year, MonthNum, MonthLabel, MonthLabelFull, AvgTemp, Visitors)

model <- lm(Visitors ~ AvgTemp, data = combined_monthly_all)

a <- coef(model)[1]
b <- coef(model)[2]
r2 <- summary(model)$r.squared

eq_text <- paste0(
  "y = ", round(b, 1), "x + (", round(a, 0), ")\\nR² = ", round(r2, 3)
)

ggplot(combined_monthly_all, aes(x = AvgTemp, y = Visitors)) +
  geom_point(color = "black", size = 2.5) +
  geom_smooth(method = "lm", color = "darkorange", se = FALSE) +
  annotate(
    "label",
    x = max(combined_monthly_all$AvgTemp, na.rm = TRUE) - 30,
    y = max(combined_monthly_all$Visitors, na.rm = TRUE) * 0.9,
    label = eq_text,
    color = "white",
    fill = "black",
    size = 4,
    label.r = unit(0.25, "lines")
  ) +
  scale_y_continuous(
    labels = scales::label_number(scale = 1e-3, suffix = "K")
  ) +
  theme_minimal(base_size = 12) +
  labs(
    title = "Visitors vs. Temperature in Acadia National Park (2015-2024)",
    x = "Average Temperature (°F)",
    y = "Monthly Visitors"
  )

```

```

)

# Visitors vs. Temperature in Acadia National Park (2015-2024)
model <- lm(AvgVisitors ~ AvgTemp, data = combined_monthly)

a <- coef(model)[1]
b <- coef(model)[2]
r2 <- summary(model)$r.squared

eq_text <- paste0(
  "y = ", round(b, 1), "x + (", round(a, 0), ")",
  "\nR² = ", round(r2, 3)
)

ggplot(combined_monthly, aes(x = AvgTemp, y = AvgVisitors, label = Month)) +
  geom_point(color = "black", size = 3) +
  geom_smooth(method = "lm", color = "darkorange", se = FALSE) +
  geom_text(vjust = -1, size = 3) +
  annotate(
    "label",
    x = max(combined_monthly$AvgTemp) - 20,
    y = max(combined_monthly$AvgVisitors) * 0.9,
    label = eq_text,
    color = "white",
    fill = "black",
    size = 4,
    label.r = unit(0.25, "lines")
  ) +
  scale_y_continuous(
    labels = scales::label_number(scale = 1e-3, suffix = "K")
  ) +
  theme_minimal() +
  labs(
    title = "Visitors vs. Temperature in Acadia National Park (2015-2024)",
    x = "Average Temperature (°F)",
    y = "Average Monthly Visitors"
  )
)

# Visitors vs. Temperature in Acadia National Park (2015-2024)
model_logistic <- nls(
  AvgVisitors ~ max_vis / (1 + exp(-k * (AvgTemp - t0))),
  data = combined_monthly,
  start = list(
    max_vis = max(combined_monthly$AvgVisitors),
    k = 0.2,
    t0 = 55
  )
)

combined_monthly$PredVisitors <- predict(model_logistic)

r2_logistic <- cor(combined_monthly$AvgVisitors, combined_monthly$PredVisitors)^2

params <- coef(model_logistic)
eq_text <- paste0(
  "y = ", round(params['max_vis'], 0),
  " / (1 + e^{-", round(params['k'], 2),
  " \cdot (x - ", round(params['t0'], 1), ")} )",
  "\nR² = ", round(r2_logistic, 3)
)

temp_seq <- seq(min(combined_monthly$AvgTemp), max(combined_monthly$AvgTemp), length.out = 200)
pred_df <- data.frame(
  AvgTemp = temp_seq,
  PredVisitors = predict(model_logistic, newdata = data.frame(AvgTemp = temp_seq))
)

ggplot() +
  geom_point(
    data = combined_monthly,
    aes(x = AvgTemp, y = AvgVisitors),
    color = "black", size = 3
  ) +
  geom_text(
    data = combined_monthly,
    aes(x = AvgTemp, y = AvgVisitors, label = Month),
    vjust = -1, size = 3
  ) +

```

```

geom_line(
  data = pred_df,
  aes(x = AvgTemp, y = PredVisitors),
  color = "darkorange", size = 1
) +
annotate(
  "label",
  x = min(combined_monthly$AvgTemp) + 18,
  y = max(combined_monthly$AvgVisitors) * 0.9,
  label = eq_text,
  color = "white",
  fill = "black",
  size = 4,
  label.r = unit(0.25, "lines"),
  family = "mono"
) +
scale_y_continuous(labels = scales::label_number(scale = 1e-3, suffix = "K")) +
theme_minimal() +
labs(
  title = "Visitors vs. Temperature in Acadia National Park (2015-2024)",
  x = "Average Temperature (°F)",
  y = "Average Monthly Visitors"
)

# === 6. EMPLOYMENT GRAPHS ===

# Hancock County Employment by Industry (2015-2024)
hancock <- employment %>%
  filter(Area == "Hancock County, Maine", Industry != "All Jobs") %>%
  arrange(Industry, Date) %>%
  group_by(Industry) %>%
  mutate(point_index = row_number()) %>%
  ungroup()

point_shapes <- c(15, 16, 17, 18, 0, 1, 2, 5, 6, 7, 8, 9)

colors_11 <- c(
  "#FF0000",
  "#FF7F00",
  "#FFFF00",
  "#7FFF00",
  "#00FF00",
  "#00FF7F",
  "#00FFFF",
  "#007FFF",
  "#0000FF",
  "#7F00FF",
  "#FF00FF"
)

ggplot(hancock, aes(x = Date, y = Employment, color = Industry)) +
  geom_line(linewidth = 0.8, alpha = 0.5) +
  geom_point(
    data = hancock %>% filter((point_index - 1) % 12 == 0),
    aes(shape = Industry),
    size = 3
  ) +
  scale_color_manual(values = colors_11) +
  scale_shape_manual(values = point_shapes) +
  theme_minimal(base_size = 14) +
  labs(
    title = "Hancock County Employment by Industry (2015-2024)",
    x = "Date",
    y = "Employment",
    color = "Industry",
    shape = "Industry"
  ) +
  theme(
    legend.position = "right",
    legend.title = element_text(face = "bold"),
    plot.title = element_text(face = "bold", size = 16)
  )

# Hancock County Elasticity of Employment Relative to Acadia Visitors

hancock_monthly <- emp_monthly_avg%>%
  filter(Area == "Hancock County, Maine", Industry != "All Jobs")

```

```

elasticity_base <- hancock_monthly %>%
  left_join(visitors_monthly_avg %>% select(Month, AvgVisitors), by = c("MonthLabel" = "Month"))

visitor_change <- elasticity_base %>%
  arrange(MonthNum) %>%
  distinct(MonthNum, AvgVisitors) %>%
  mutate(
    pct_visitors = (AvgVisitors - lag(AvgVisitors)) / lag(AvgVisitors) * 100
  ) %>%
  mutate(
    pct_visitors = ifelse(
      is.na(pct_visitors),
      (AvgVisitors[MonthNum == 1] - AvgVisitors[MonthNum == 12]) /
      AvgVisitors[MonthNum == 12] * 100,
      pct_visitors
    )
  )

elasticity_base <- elasticity_base %>%
  left_join(visitor_change %>% select(MonthNum, pct_visitors),
    by = "MonthNum")

elasticity_base <- elasticity_base %>%
  arrange(Industry, MonthNum) %>%
  group_by(Industry) %>%
  mutate(
    pct_employment =
      (AvgEmployment - lag(AvgEmployment)) / lag(AvgEmployment) * 100,
    pct_employment = ifelse(
      is.na(pct_employment),
      (AvgEmployment[MonthNum == 1] - AvgEmployment[MonthNum == 12]) /
      AvgEmployment[MonthNum == 12] * 100,
      pct_employment
    )
  ) %>%
  ungroup()

elasticity_base <- elasticity_base %>%
  mutate(
    elasticity = pct_employment / pct_visitors,
    MonthLabel = factor(MonthLabel, levels = month.abb, ordered = TRUE)
  )

ggplot(elasticity_base,
  aes(x = MonthLabel, y = elasticity, color = Industry)) +

  geom_line(aes(group = Industry), linewidth = 1, alpha = 0.6) +
  geom_point(aes(shape = Industry), size = 3) +

  scale_color_manual(values = colors_11) +
  scale_shape_manual(values = point_shapes) +

  theme_minimal(base_size = 14) +
  labs(
    title = "Hancock County Elasticity of Employment Relative to Acadia Visitors",
    x = "Month",
    y = "Elasticity",
    color = "Industry",
    shape = "Industry"
  ) +
  theme(
    legend.position = "right",
    legend.title = element_text(face = "bold"),
    plot.title = element_text(face = "bold", size = 16),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

# Hancock County Seasonality Contribution Index (SCI) by Industry (Peak Season)

industry_correlation <- elasticity_base %>%
  group_by(Industry) %>%
  summarize(
    correlation = cor(AvgEmployment, AvgVisitors, use = "complete.obs")
  )

peak_season <- elasticity_base %>%
  filter(MonthLabel %in% c("Jul", "Aug", "Sep")) %>%

```

```

group_by(Industry) %>%
  summarize(peak_emp = sum(AvgEmployment)) %>%
  ungroup() %>%
  mutate(
    peak_share = peak_emp / sum(peak_emp)
  )

SCI_peak <- industry_correlation %>%
  left_join(peak_season, by = "Industry") %>%
  mutate(SCI_peak = abs(correlation) * peak_share) %>%
  arrange(desc(SCI_peak))

SCI_peak <- SCI_peak %>%
  arrange(desc(SCI_peak)) %>%
  mutate(Industry = factor(Industry, levels = rev(Industry)))

ggplot(SCI_peak, aes(x = Industry, y = SCI_peak, fill = SCI_peak)) +
  geom_col() +
  geom_text(
    aes(label = sprintf("%.3f", SCI_peak)),
    hjust = -0.1,
    size = 4
  ) +
  coord_flip() +
  scale_y_continuous(
    breaks = seq(0, max(SCI_peak$SCI_peak) * 1.1, by = 0.05),
    labels = function(x) sprintf("%.2f", x)
  ) +
  scale_fill_gradient(low = "lightblue", high = "darkblue") +
  theme_minimal(base_size = 14) +
  labs(
    title = "Hancock County Seasonality Contribution Index (SCI) by Industry (Peak Season)",
    x = "Industry",
    y = "SCI (Correlation x Peak Season Share)",
    fill = "SCI"
  ) +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    axis.text.y = element_text(face = "bold")
  ) +
  expand_limits(x = max(SCI_peak$SCI_peak) * 1.25)

# Employment by Industry and County (2015-2024)

top3 <- c(
  "Professional and business services",
  "Leisure and hospitality",
  "Trade, transportation, and utilities"
)

df <- employment %>%
  filter(Industry %in% c("All Jobs", top3)) %>%
  group_by(Area, Date) %>%
  mutate(AllJobs = Employment[Industry == "All Jobs"]) %>%
  ungroup()

everything_else <- employment %>%
  filter(!(Industry %in% c("All Jobs", top3))) %>%
  group_by(Area, Date, Tier, Approx_Distance_miles) %>%
  summarize(Employment = sum(Employment, na.rm = TRUE), .groups = "drop") %>%
  mutate(Industry = "Excluding Top 3 Industries")

plot_df <- df %>%
  select(Area, Tier, Approx_Distance_miles, Date, Industry, Employment) %>%
  bind_rows(everything_else)

plot_df <- plot_df %>%
  mutate(
    FacetLabel = glue("{Area}\n({Approx_Distance_miles} miles from Acadia)")
  )

plot_df$FacetLabel <- fct_reorder(plot_df$FacetLabel, plot_df$Tier, .fun = min)

plot_df <- plot_df %>%
  mutate(Industry = ifelse(Industry == "All Jobs", "All Industries", Industry))

color_map <- c(
  "All Industries" = "black",

```

```

"Professional and business services" = "#007FFF",
"Leisure and hospitality" = "#FF0000",
"Trade, transportation, and utilities" = "#00AA00",
"Excluding Top 3 Industries" = "#FFA500"
)

ggplot(plot_df, aes(x = Date, y = Employment, color = Industry)) +
  geom_line(size = 1) +
  facet_wrap(~ FacetLabel, ncol = 2, scales = "free_y") +
  scale_y_continuous(expand = expansion(mult = c(0, .05)), limits = c(0, NA)) +
  scale_color_manual(values = color_map) +
  labs(
    title = "Employment by Industry and County (2015-2024)",
    x = "Date",
    y = "Employment",
    color = "Industry"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "bottom",
    strip.text = element_text(size = 12, face = "bold")
  )

# Employment and Monthly Acadia Visitor Trends (2015-2024)

emp_wide <- emp_monthly_avg %>%
  filter(Industry %in% c("All Jobs", top3)) %>%
  select(Area, Tier, Approx_Distance_miles, Industry, MonthNum, MonthLabel, AvgEmployment) %>%
  pivot_wider(
    names_from = Industry,
    values_from = AvgEmployment
  ) %>%
  mutate(
    `Excluding Top 3 Industries` = `All Jobs` -
    `Leisure and hospitality` -
    `Professional and business services` -
    `Trade, transportation, and utilities`
  )

plot_data <- emp_wide %>%
  pivot_longer(
    cols = c(`All Jobs`,
    `Leisure and hospitality`,
    `Professional and business services`,
    `Trade, transportation, and utilities`,
    `Excluding Top 3 Industries`),
    names_to = "EmploymentType",
    values_to = "Employment"
  )

visitors_aligned <- visitors_monthly_avg %>%
  mutate(MonthNum = match(Month, month.abb))

plot_data <- plot_data %>%
  left_join(visitors_aligned %>% select(MonthNum, AvgVisitors),
    by = "MonthNum")

plot_data <- plot_data %>%
  mutate(
    FacetLabel = glue("{Area}\n({Approx_Distance_miles} miles from Acadia)")
  )

plot_data$FacetLabel <- fct_reorder(plot_data$FacetLabel, plot_data$Tier, min)

plot_data <- plot_data %>%
  group_by(FacetLabel) %>%
  mutate(
    MaxEmp = max(Employment, na.rm = TRUE),
    VisitorShape = (AvgVisitors / max(AvgVisitors, na.rm = TRUE)) * MaxEmp
  ) %>%
  ungroup()

plot_data <- plot_data %>%
  mutate(EmploymentType = ifelse(EmploymentType == "All Jobs", "All Industries", EmploymentType))

ggplot(plot_data, aes(x = MonthNum, y = Employment, color = EmploymentType)) +
  geom_line(size = 1.1) +
  geom_point(size = 1.5) +

```

```

geom_line(aes(y = VisitorShape),
  color = "black", linetype = "dashed", size = 1) +
geom_point(aes(y = VisitorShape),
  color = "black", size = 2) +
scale_x_continuous(breaks = 1:12, labels = month.abb) +
scale_color_manual(values = color_map) +
facet_wrap(~ FacetLabel, ncol = 2, scales = "free_y") +
labs(
  title = "Employment and Monthly Acadia Visitor Trends (2015-2024)",
  x = "Month",
  y = "Employment",
  color = "Employment Type"
) +
theme_minimal(base_size = 14) +
theme(
  legend.position = "bottom",
  strip.text = element_text(size = 12, face = "bold"),
  axis.text.x = element_text(size = 10)
)

# Hancock County Seasonal Employment Increase from Low to Peak Season (2015-2024)

all_jobs <- hancock_monthly %>%
  group_by(MonthNum, MonthLabel) %>%
  summarize(
    AvgEmployment = sum(AvgEmployment, na.rm = TRUE),
    Industry = "All Industries",
    .groups = "drop"
  )

combined <- bind_rows(hancock_monthly, all_jobs)

jan_aug <- combined %>%
  filter(MonthNum %in% c(1, 8)) %>%
  select(Industry, MonthNum, AvgEmployment) %>%
  pivot_wider(
    names_from = MonthNum,
    values_from = AvgEmployment,
    names_prefix = "Month_"
  ) %>%
  mutate(
    Change = Month_8 - Month_1
  )

total_change <- jan_aug$Change[jan_aug$Industry == "All Industries"]

jan_aug <- jan_aug %>%
  mutate(
    PercentOfTotal = ifelse(
      Industry != "All Industries",
      round(100 * Change / total_change, 1),
      NA
    ),
    Label = ifelse(
      Industry == "All Industries",
      paste0("Δ = ", Change),
      paste0("Δ = ", Change, " (" , PercentOfTotal, "% of Total Δ)")
    )
  )

colors_12 <- c(
  "All Industries" = "black",
  "Construction" = "#FF0000",
  "Education and health services" = "#FF7F00",
  "Financial activities" = "#FFFF00",
  "Government" = "#7FFF00",
  "Information" = "#00FF00",
  "Leisure and hospitality" = "#00FF7F",
  "Manufacturing" = "#00FFFF",
  "Natural resources and mining" = "#007FFF",
  "Other services" = "#0000FF",
  "Professional and business services" = "#7F00FF",
  "Trade, transportation, and utilities" = "#FF00FF"
)

ggplot(jan_aug, aes(x = reorder(Industry, Change), y = Change, fill = Industry)) +
  geom_col() +
  coord_flip() +

```

```

geom_text(aes(label = label), hjust = -0.1, size = 3.5) +
scale_fill_manual(values = colors_12) +
labs(
  title = "Hancock County Average Seasonal Employment Increase from Low to Peak Season (2015-2024)",
  subtitle = "Low Season = January & Peak Season = August",
  x = "",
  y = "Employment Change"
) +
theme_minimal(base_size = 14) +
theme(
  legend.position = "none",
  axis.text.y = element_text(face = "bold"),
  plot.title = element_text(face = "bold", size = 16),
  plot.subtitle = element_text(size = 12)
) +
scale_y_continuous(expand = expansion(mult = c(0, 0.15)))

# === 7. FINAL ANALYSIS ===

# Hancock County Monthly Employment vs. Acadia National Park Monthly Visitors (2015-2024)
hancock <- emp_monthly_avg %>%
  filter(Area == "Hancock County, Maine")

everything_else <- hancock %>%
  filter(!(Industry %in% c("All Jobs", top3))) %>%
  group_by(MonthNum, MonthLabel) %>%
  summarize(
    AvgEmployment = sum(AvgEmployment, na.rm = TRUE),
    Industry = "Excluding Top 3 Industries",
    .groups = "drop"
  )

hancock_plot <- hancock %>%
  filter(Industry %in% c("All Jobs", top3)) %>%
  bind_rows(everything_else) %>%
  left_join(visitors_monthly_avg, by = c("MonthLabel" = "Month"))

eq_labels <- hancock_plot %>%
  group_by(Industry) %>%
  summarise(
    lm_fit = list(lm(AvgEmployment ~ AvgVisitors)),
    .groups = "drop"
  ) %>%
  rowwise() %>%
  mutate(
    slope = coef(lm_fit)[2],
    intercept = coef(lm_fit)[1],
    r2 = summary(lm_fit)$r.squared,
    label = paste0("y = ", round(intercept, 2),
      " + ", round(slope, 5), "x\nR² = ", round(r2, 2)),
    x_pos = max(hancock_plot$AvgVisitors[hancock_plot$Industry == Industry]) + 100000,
    y_pos = max(hancock_plot$AvgEmployment[hancock_plot$Industry == Industry]) + 300
  )

ggplot(hancock_plot, aes(x = AvgVisitors, y = AvgEmployment, color = ifelse(Industry == "All Jobs", "All Industries", Industry))) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", se = FALSE, formula = y ~ x) +
  geom_text(
    data = eq_labels,
    aes(
      x = x_pos,
      y = y_pos,
      label = label,
      color = ifelse(Industry == "All Jobs", "All Industries", Industry)
    ),
    hjust = 1, size = 4,
    show.legend = FALSE
  ) +
  scale_color_manual(values = color_map) +
  scale_x_continuous(labels = scales::label_number(scale = 1e-3, suffix = "K")) +
  scale_y_continuous(labels = scales::label_number(scale = 1e-3, suffix = "K")) +
  labs(
    title = "Hancock County Monthly Employment vs. Acadia National Park Monthly Visitors (2015-2024)",
    x = "Average Monthly Visitors",
    y = "Average Monthly Employment",
    color = "Industry"
  ) +

```

```

theme_minimal(base_size = 14) +
theme(
  legend.position = "bottom"
)

# Interpreting the slopes
slopes_table <- eq_labels %>%
  mutate(
    Industry = ifelse(Industry == "All Jobs", "All Industries", Industry),
    InverseSlope = 1 / slope
  ) %>%
  select(Industry, slope, InverseSlope, r2)

slopes_table_plot <- slopes_table %>%
  select(Industry, slope, InverseSlope) %>%
  mutate(
    slope = round(slope, 6),
    InverseSlope = round(InverseSlope)
  ) %>%
  gt() %>%
  tab_header(
    title = "Employment Sensitivity to Park Visitation",
    subtitle = "Slope and Inverse Slope of Employment ~ Visitors Regression"
  ) %>%
  cols_label(
    Industry = "Industry",
    slope = "Slope",
    InverseSlope = "Inverse Slope"
  ) %>%
  fmt_number(
    columns = slope,
    decimals = 5
  ) %>%
  fmt_number(
    columns = InverseSlope,
    decimals = 0
  ) %>%
  tab_options(
    table.font.size = 14,
    heading.align = "center"
  )

```